

Session 17:

AI That Combines Everything - Multimodal AI

Today, you'll discover how AI can see, hear, and read all at once. We'll explore how AI fuses information from different senses to understand the world more completely. You'll use tools that analyze images, generate videos from text, and even create impossible scenes - all while bringing the Sense → Think → Act framework full circle.



Objectives

By the end of this session, you will be able to:

01

Define Multimodal AI

Explain how it integrates text, image, audio, and video.

02

Apply Sense → Think → Act

Use the framework to analyze multimodal systems.

03

Describe Multimodal Processing

Explain how multimodal models process and connect different types of data.

04

Identify Applications & Challenges

Recognize real-world uses and limitations of multimodal AI.

05

Create a Multimodal Output

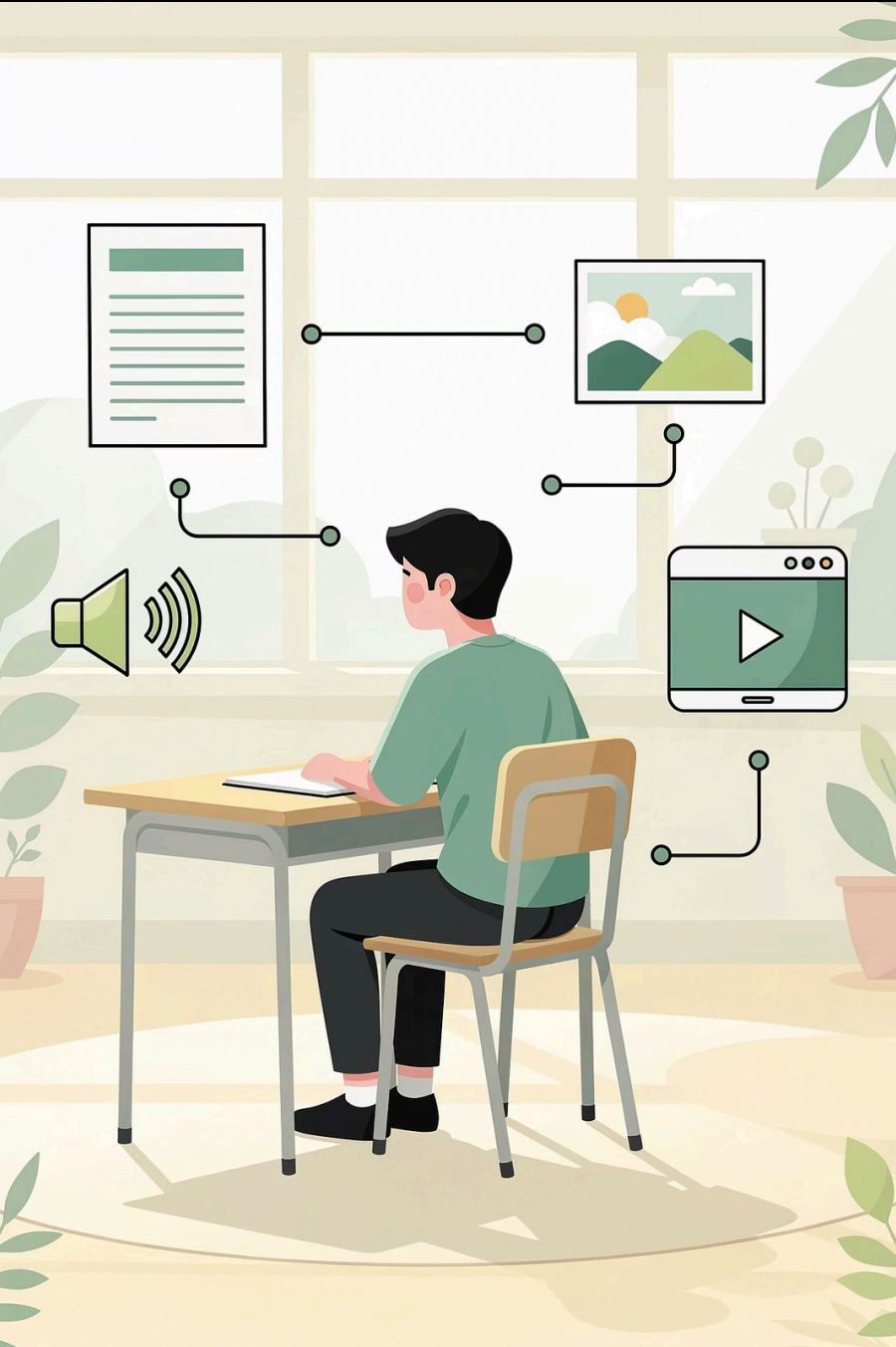
Use a text-to-video tool to produce your own multimodal creation.

Recap: Sessions 14, 15 & 16

Let's revisit the perception frameworks you've already mastered.

Domain	Framework	What It Does
NLP (Session 14)	Break → Understand → Generate	AI reads and writes text.
Computer Vision (Session 15)	See → Process → Act	AI interprets images and video.
Voice AI (Session 16)	Hear → Process → Speak	AI listens and speaks.

 **Key Insight:** Each of these frameworks is a specialized version of the same foundational idea introduced in Session 1: **Sense → Think → Act**.



What is Multimodal AI?

Multimodal AI is artificial intelligence that processes and reasons about multiple types of data at once - text, images, audio, video, etc.

When you watch a movie, you see the action, hear the dialogue, and read the subtitles all at the same time. Your brain combines all that information to understand the story. **Multimodal AI tries to do the same thing.**

Unimodal AI

A model that only reads text, or only analyzes images. It has one sense.

Example: A spam filter only reads email text. It doesn't know if there's a funny picture attached.

Multimodal AI

A model that can examine a photo *and* read a caption *and* listen to audio - all together. It has **multiple senses**.

Example: A smart assistant that looks at a photo of a dish, reads the recipe request, and speaks the instructions aloud.

Unimodal vs. Multimodal AI

	Unimodal AI	Multimodal AI
Data It Uses	One type (e.g., only text)	Multiple types (text + image + audio)
Understanding	Limited to one perspective	Fuller pattern matching, not true understanding
Example	Spam filter reading emails	A smart assistant looking at a photo, listening to a voice, and reading comments



Example

Photo: A hedgehog wearing a tiny raincoat. **Prompt:** "Write a clever caption."

1

Translate Image

The AI translates the image into a shared number code.

2

Translate Text

The AI translates the text into the same code.

3

Fuse Clues

Raincoat = wet weather prep. Hedgehog in human clothing = funny.

4

Result

"Even hedgehogs check the weather forecast now."

This integrated understanding - connecting visual details with a creative request is far more powerful than any single-mode AI. **But it's all math, not magic.**

How Multimodal AI Works:

Sense → Think → Act Across Modalities

Most multimodal AI systems follow the same three-stage process you've known since Session 1. This framework helps you see what's happening inside when AI combines multiple senses.

Stage	What Happens in Multimodal AI	Hedgehog Photo Example
SENSE	AI collects raw data from several sources at once.	Pixels from the hedgehog image + tokens from your text prompt.
THINK	AI fuses information across modalities to build a unified interpretation.	Recognizes "hedgehog" + "raincoat" + understands the request for a "clever caption."
ACT	AI produces an output which may itself be multimodal.	Generates a text caption; could also narrate the description aloud.

 **Important:** These stages happen in milliseconds. The AI doesn't "understand" like a human - it follows mathematical steps to find and connect patterns.

Stage 1: SENSE - Gathering Multiple Streams of Data

Before AI can work with different types of information, it must first collect them. This is the **Sense** stage.

Input Type	What the AI Collects	Example
Text	Sequences of characters and words.	"Write a clever caption."
Images	Grids of pixels, each with Red, Green, Blue values.	A photo of a hedgehog in a raincoat → millions of colored dots.
Audio	Sound waves converted into digital waveforms (amplitude over time).	A voice saying "I'm fine" → a wavy line of numbers.

 **Key Insight:** At this stage, the data is still in its raw, native format.

Stage 2: THINK - Encoding, Aligning, and Fusing

Encode

Specialized encoders turn each type of raw data - whether pixels from an image, sound waves from audio, or words from text into numerical representations that capture meaning. These lists of numbers are called **vector embeddings**.

***Example:** Converting different file formats (JPEG image, MP3 audio, TXT document) into the same digital language that a computer can process uniformly.*

Align

All the vector embeddings are placed into a shared mathematical space called a **Joint Embedding Space**. Through training, the model learns to position semantically similar concepts close together, regardless of whether they came from text, image, or audio.

***Example:** Organizing a library where all materials about a specific topic share the same shelf location - whether they are books, audiobooks, or documentary films.*

Fuse

The AI combines the aligned embeddings from different modalities - using techniques like attention mechanisms that weigh the importance of each piece of information to form a single, rich representation of the whole situation.

***Example:** A chef combining ingredients from different recipes (pasta from one, sauce from another, seasoning from a third) to create an entirely new dish.*

Stage 3: ACT - Generating Multimodal Outputs

Once the AI has a fused mathematical representation, it can produce an output. Because it works with multiple modalities, it can act in many formats.

Text

Captions, answers, summaries, stories.

"Ready for the desert fiesta."

Image

New images from text prompts.

"A teapot shaped like an elephant pouring steaming tea into a tiny cup."

Video

New video clips from descriptions.

"A grandfather clock dancing the tango with a floor lamp in an empty ballroom."

Audio

Synthesized speech from text.

A voice narrating the story you wrote.

Cross-Modal Retrieval

Finding one modality using another.

Searching for images by typing "cactus in a sombrero."



Putting It All Together: A Complete Example

Scenario: You type a text prompt into a multimodal AI: *"Generate an image of a flamingo wearing a detective hat and holding a magnifying glass."*



SENSE

Text prompt →
tokens: "flamingo,"
"detective hat,"
"magnifying glass."



THINK

Encode → text
embedding.
Align → lands near
"flamingo," "detective
hat," "magnifying
glass."
Fuse → single scene
representation.



ACT

AI generates a brand
new image: a
flamingo standing
tall, wearing a tiny
detective hat, and
holding a magnifying
glass in its beak.

Mini Activity 1: Applying the Multimodal Framework

Instructions: Read each scenario and answer the question by applying what you've learned about how multimodal AI works. Don't just guess - think through the stages.

Scenario	Your Answer
1 You ask a multimodal AI to describe a photo of a raccoon holding a tiny umbrella. The AI says, "A raccoon holds an umbrella." What stages did the AI go through to produce this answer? List them in order.	
2 Why does the AI need an "Align" step? What would happen if it skipped alignment and tried to fuse the codes directly?	
3 An AI generates an image of "a sloth wearing a party hat and holding a balloon." The image shows a sloth and a party hat, but the balloon is missing. What likely went wrong in the AI's process?	

Mini Activity 1: Answer Key

1 Sense → Think (Encode → Align → Fuse) → Act

The AI must first sense the image and text, encode both into secret codes, align them so "raccoon" and "umbrella" map nearby, fuse that aligned info, and finally act by generating text.

2 Alignment is Essential

Alignment ensures that the secret codes for similar concepts are close together in the Joint Embedding Space. Without it, the code for "raccoon" from the image and the code for "raccoon" from the text might be completely unrelated, and the AI wouldn't know they refer to the same thing. Fusion would produce nonsense.

3 Partial Fusion Failure

The AI successfully encoded and aligned the sloth and hat but likely failed to fuse the balloon concept correctly. It understood some nouns but not all of them. This is a common failure - some visual concepts are harder to represent together than others.

Common Multimodal AI Tasks

Task	What It Does	Example
Image Captioning	Generates a text description of an image.	Photo of a lighthouse with a glowing orb floating above it → <i>"A lighthouse stands beneath a mysterious glowing orb in the night sky."</i>
Visual Question Answering (VQA)	Answers specific questions about an image.	<i>"What is floating above the lighthouse?"</i> → <i>"A glowing orb."</i>
Text-to-Image Generation	Creates a new image from a text prompt.	<i>"An hourglass with swirling galaxies inside the sand."</i> → AI generates the image.

Common Multimodal AI Tasks

Task	What It Does	Example
Text-to-Video Generation	Creates a new video clip from a text prompt.	<i>"A typewriter typing by itself on a desk in a dim room, sheets of paper rising gently into the air."</i> → AI generates the clip.
Speech Recognition (ASR/STT)	Converts spoken audio to text.	Voice memo → <i>"Order two pizzas and a side salad."</i>
Text-to-Speech (TTS)	Converts written text to spoken audio.	<i>"Your package will arrive tomorrow morning."</i> → Spoken aloud.
Cross-Modal Retrieval	Finds one modality using another.	Search <i>"lighthouse with glowing orb"</i> → Returns matching photos.

Challenges in Multimodal AI

Even powerful systems have limits.

Alignment

Modalities must be perfectly synchronized. Audio and video out of sync confuse the AI.

Deepfakes

Malicious use to fabricate convincing fake media. A video of someone saying words they never spoke.

Computational Cost

Processing multiple modalities requires massive resources. Multimodal models are larger and slower.

Cross-Modal Bias

Bias can spread across text, image, and video data. A model may reinforce stereotypes in captions and visuals.

Data Scarcity

High-quality paired datasets are expensive to create. Models may struggle with rare modality combinations.

 **Remember:** Human oversight is essential. Verify outputs, especially when they seem flawless.



Mini Activity 2 : Try Creative Video Prompts

Here's what video AI can imagine from these prompts:

Prompt	What Might Be Generated
<i>"A young goat in sunglasses dancing on a grand piano."</i>	A playful clip of a goat hopping across piano keys.
<i>"A slice of pizza drifting through space near Saturn's rings."</i>	A cinematic shot of floating pizza against a starry backdrop.
<i>"A robot tending a field of glowing blue flowers at dusk."</i>	A futuristic scene with gentle motion.

Real-World Multimodal AI Applications

Multimodal AI solves real problems.



Autonomous Vehicles

Sense: Cameras, radar, lidar.

Think: Fuse sensor data to detect lanes, signs, obstacles.

Act: Steer, accelerate, brake.



Accessibility

Sense: Image or text.

Think: Recognize objects and context.

Act: Generate spoken alt-text for visually impaired users.



Content Moderation

Sense: Text, audio, images.

Think: Identify policy violations across modalities.

Act: Flag or remove harmful material.



Medical Diagnosis

Sense: Medical scans + patient notes.

Think: Correlate visual findings with textual history.

Act: Highlight potential concerns.



Retail Search

Sense: Text query + uploaded photo. **Think:** Fuse visual style with keywords. **Act:** Display matching products.



Hands-On Activity: The Multimodal Creative Lab

Goal: Use multimodal AI tools as creative partners. You supply the raw ideas. AI helps bring them to life.



**Multimodal AI
Chat**

e.g., Google
Gemini



**Video
Generation Tool**

e.g., Google Veo 3
or similar



**Text-to-Video
Tool**

e.g., Pictory

Part 1: Work with Your Own Image



Use a photo you have

Take a picture of something nearby: a toy, a pet, a plant, a book cover, or an interesting object in your room.



Find an image online

Search for a free-to-use image (e.g., "funny animal photo" or "unusual object").



Generate an image with Gemini

If you don't have an image, ask Gemini to create one! Type a prompt like: "Generate an image of a curious squirrel wearing a tiny backpack." Once generated, use that image for the activity.

Part 2: Imagine the Impossible with Gemini Veo 3

Step 1: Build an Impossible Scene. Combine one item from each column (or invent your own).

Character	Action	Setting
A penguin in a tuxedo	Playing a flaming electric guitar	Inside a snow globe
A giant talking mushroom	Riding a unicycle on a rainbow	On the rings of Saturn
A robot made of balloons	Having a tea party underwater	In a library where books fly

Your Scene (one sentence): *"A [character] [action] [setting]."*

Step 2: Generate the Video. Type your scene description into a video AI tool and watch what it creates.

Part 3: Film Creative

Write a Three-Sentence Script. Pick a starter or write your own.

Starter	First Sentence
A	<i>"Marvin the mushroom had always dreamed of seeing the ocean."</i>
B	<i>"The old vending machine in the hallway was actually a portal."</i>
C	<i>"Nobody believed that Toast, the cat, understood every word."</i>
Your Own	[Write your own opening line.]

Open Text-to-Video Tool

Click **"Script to Video"** and paste your three sentences.

Choose a Template

Select a template that fits the mood of your story.

Make One Creative Edit

Swap a scene for something unexpected, then generate your film.

Summary

Concept	Key Idea
Multimodal AI	AI that processes text, images, audio, video, and code together.
Modality	A single type of data (text, image, sound, video, code).
Fusion	Combining clues from different modalities for deeper insight.
Sense → Think → Act	The foundational framework - now applied across multiple senses.

Summary

Concept	Key Idea
Multimodal Models	Can mix inputs: image + text, video + text, audio + text, code + diagrams.
Text-to-Video AI	Senses text; thinks by matching visuals; acts by generating a video.
Video AI	Senses video or text; thinks across time or imagination; acts with descriptions or new video.

Key Takeaway

Multimodal AI = Sense → Think → Act

SENSE

Gather text, image, audio, video, code - any combination.

THINK

Translate all senses into a shared internal language; fuse clues for deeper understanding.

ACT

Generate text, descriptions, videos, code, or coordinated actions.

Cross-Modal Understanding

Connect information across different input types.

Mixed-Input Reasoning

Answer questions that require combining multiple sources.

Multimodal Generation

Create new content in one format from another.

Homework: Make a Mini-Movie

Get a Story Idea (3 min)

Open **Gemini** and ask: *"Give me a short three-sentence story about an unusual animal on a small adventure."* Copy the story.



Create One Scene (3 min)

Open **Veo 3** (or any video tool). Type a description of the most exciting moment from your story. Example: *"A small glowing mushroom in a dark forest with a curious mouse nearby."* Generate the short clip.



Make the Movie (7 min)

Open **Pictory** and click **"Script to Video."** Paste your three-sentence story. Choose any template. If possible, add your Veo 3 clip. Generate and save your video.

Next Session Preview

Session 18 : AI in Everyday Life & Across Industries

Now that you understand how AI integrates multiple senses, we'll explore how these capabilities are reshaping healthcare, finance, entertainment, education, transportation, etc. Get ready to see AI's impact everywhere.

